



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2020 - 2021)

JEE(Advanced)
REVIEW TEST
13-06-2021

JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : ALL STAR BATCHES)

PAPER-1

Time : 3 Hours

Maximum Marks : 183

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **28** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140,

- | | |
|------------------------------------|--|
| • Boltzmann constant | $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$ |
| • Coulomb's law constant | $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ |
| • Universal gravitational constant | $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$ |
| • Speed of light in vacuum | $c = 3 \times 10^8 \text{ ms}^{-1}$ |
| • Stefan-Boltzmann constant | $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$ |
| • Wien's displacement law constant | $b = 2.89 \times 10^{-3} \text{ m-K}$ |
| • Permeability of vacuum | $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$ |
| • Permittivity of vacuum | $\epsilon_0 = \frac{1}{\mu_0 c^2}$ |
| • Planck constant | $h = 6.63 \times 10^{-34} \text{ J-s}$ |

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS**BEWARE OF NEGATIVE MARKING****PART-1 : PHYSICS****SECTION-I(i) : (Maximum Marks: 28)**

- This section contains **SEVEN** questions.
 - Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
 - For each question, choose the correct option(s) to answer the question.
 - Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -1 In all other cases.
 - **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.
-
1. Regarding photoelectric effect which of the following statement(s) is/are **CORRECT**.
- (A) Within the limit of experimental accuracy, there is no time interval between the arrival of light at a metal surface and emission of photoelectrons.
- (B) A bright light yields more photoelectrons than a dim one of same frequency but energy of fastest electrons remain the same.
- (C) Higher the frequency of light, more is the energy of photoelectrons
- (D) At frequencies below a certain critical frequency, which is characteristic of each particular metal, no electrons are emitted.

Space for Rough Work

2. Choose **CORRECT** statement(s) :-

- (A) The binding energy per nucleon for a given nucleus is an average found by dividing its total binding energy by the number of nucleons it contains
- (B) Energy needed to remove a neutron from $^{42}_{20}\text{Ca}$ is same as to remove a proton from $^{42}_{20}\text{Ca}$
- (C) The helium isotope ^6_2He is unstable. ^6_2He can undergo negative beta decay to become lithium isotope ^6_3Li
- (D) Energy needed to remove a neutron from $^{42}_{20}\text{Ca}$ is different from the energy needed to remove a proton from $^{42}_{20}\text{Ca}$

3. A thin equiconvex lens of refractive index $\frac{3}{2}$ is placed on a horizontal plane mirror as given

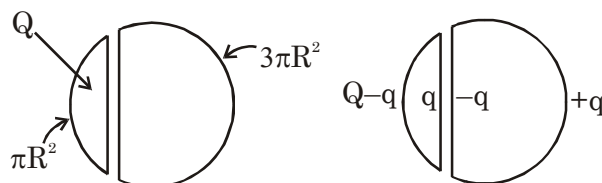
in diagram. The space between lens and mirror is filled with water of refractive index $\frac{4}{3}$. It is found that when a point object is placed 15 cm above the lens on its principal axis, object coincides, with its own image. Choose **CORRECT** statement(s).



- (A) Radius of curvature of convex surface of lens is 15 cm
- (B) Radius of curvature of convex surface of lens is 10 cm
- (C) Focal length of equiconvex lens in air is 15 cm
- (D) Focal length of equiconvex lens in air is 10 cm

Space for Rough Work

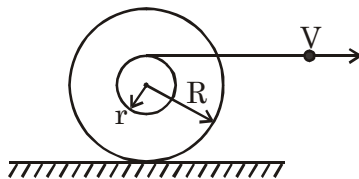
4. A solid metal sphere of radius r is divided into two parts by a planer cut, made in such a way that the outer surface of smaller part of the sphere has area πR^2 . The cut surfaces are coated with negligibly thin insulating layer, and the two parts are put together again. so that original shape of sphere is resorted. Initially sphere is neutral. The smaller part is given a charge $+Q$ while the larger part is neutral. Choose **CORRECT** statement(s) :



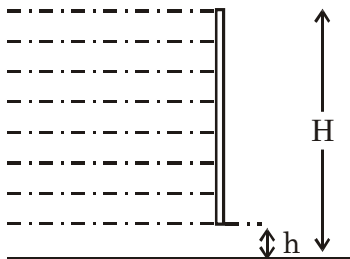
- (A) The insulated cut between the two parts becomes a parallel plate capacitor, with negligible energy.
- (B) If we denote the charge q on the capacitor corresponding to insulated space, then total charge on curved surface of entire sphere is $+Q$ which is independent of value of q .
- (C) Charge distribution on the curved surface of the sphere is uniform
- (D) Electric field at a point at a distance $r(>R)$ from the centre of sphere is $E = \frac{Q}{4\pi\epsilon_0 r^2}$

Space for Rough Work

5. A reel is being pulled along a horizontal surface by a string wrapped around as shown. There is no slipping between the reel and the surface. The string is pulled at a constant speed v .



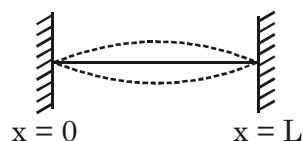
- (A) Angular speed of reel is $\omega = \frac{V}{R+r}$ (B) Speed of centre of reel is $\frac{VR}{R+r}$
- (C) Angular speed of reel is $\omega = \frac{V}{R-r}$ (D) Speed of centre of reel is $\frac{VR}{R-r}$
6. Water is filled upto height H in a dam. In one wall of dam, a gate of height h ($\ll H$) and length l is opened. Neglect viscosity



- (A) Velocity of liquid coming out of gate is $v = \sqrt{2gH}$
- (B) Rate of flow of liquid coming out of gate is $Q = \ell h \sqrt{2gH}$
- (C) Velocity of liquid coming out of gate is $v = \sqrt{gH}$
- (D) Rate of flow of liquid coming out of gate is $Q = \ell h \sqrt{gH}$

Space for Rough Work

7. A string fixed at both ends is stretched between $x = 0$ and $x = L$. String is vibrating in fundamental mode. Choose **CORRECT** statement(s) :



- (A) Power transmission through a point at $x = \frac{L}{2}$ is zero at all times
- (B) Total energy of string does not change with time (ignore losses)
- (C) For a small element of length dx at location $x = \frac{L}{4}$, total energy changes with time. (ignore losses)
- (D) For a small element of length dx at location $x = \frac{L}{4}$, total energy does not change with time

Space for Rough Work

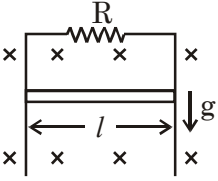
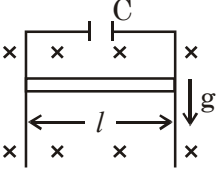
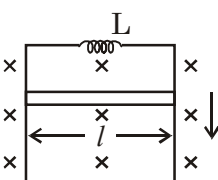
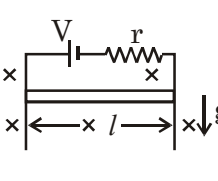
SECTION-I(ii) : (Maximum Marks : 18)

- This section contains **THREE** questions of matching type.
- This section contains **TWO** table (having 3 columns and 4 rows)
- Based on table, there are **THREE** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct option is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

Answer Q.8, Q.9 and Q.10 by appropriately matching the information given in the three columns of the following table.

Three columns are given. In column I all situations are in vertical plane. In all cases conductor is released from rest in a uniform horizontal magnetic field B inward.

Conductor has mass m and is having negligible resistance. Frictional force be ignored.

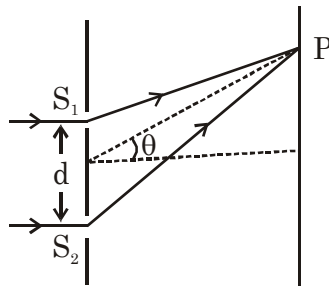
Column-I	Column-II	Column-III
(i) 	(I) Conductor may remain at rest	(P) Conductor will perform simple harmonic
(ii) 	(II) Conductor will fall with decreasing acceleration, which eventually will become zero	(Q) Conductor will attain terminal velocity
(iii) 	(III) Conductor will fall with constant acceleration	(R) Conductor will undergo uniformly accelerated motion
(iv) 	(IV) Conductor will fall with variable acceleration. Variable acceleration will vary harmonically	(S) First conductor will not move then after some time it will perform oscillation

8. Which of the following is correct?
 (A) (i), III, R (B) (ii), III, R (C) (i), II, P (D) (iii), I, P
9. Which of the following is correct?
 (A) (i), II, Q (B) (i), III, Q (C) (ii), II, Q (D) (iii), I, Q
10. Which of the following is correct?
 (A) (ii), III, S (B) (iii), I, Q (C) (ii), IV, P (D) (iii), IV, P

Space for Rough Work

Answer Q.11, Q.12 and Q.13 by appropriately matching the information given in the three columns of the following table.

Consider the arrangement given in the diagram. This set up correspond to standard Young's double slit experiment. Wavelength of light used is λ . Assume that at point P, intensity produced by both beams is same and is equal to I_0 .



Column-I	Column-II (Value of angle θ)	Column III (Resultant intensity)
(i) $d = 2\lambda$	(I) $\theta = \frac{\pi}{6}$	(P) 0
(ii) $d = \frac{3\lambda}{2}$	(II) $\theta = \sin^{-1} \frac{1}{4}$	(Q) $3I_0$
(iii) $d = \lambda$	(III) $\theta = \sin^{-1} \frac{1}{3}$	(R) $2I_0$
(iv) $d = 1.2\lambda$	(IV) $\theta = \sin^{-1} \frac{1}{6}$	(S) I_0
11. Which of the following is correct? (A) (i), I, P (B) (i), II, Q (C) (i), II, P (D) (i), IV, P		
12. Which of the following is correct? (A) (ii), I, P (B) (ii), I, R (C) (ii), II, R (D) (ii), IV, R		
13. Which of the following is correct? (A) (iii), IV, Q (B) (iii), III, R (C) (iii), III, P (D) (iii), III, Q		

Space for Rough Work

SECTION-II : (Maximum Marks : 15)

- This section contains **FIVE** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

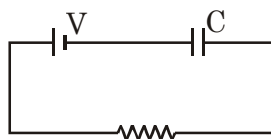
+		-	
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

1. A helicopter can hover when the power output of engine is P . Let us assume power P depends on linear size L , average density of helicopter ρ_{hel} , density of air ρ_{air} and acceleration due to gravity g . so power $P \propto g^\alpha L^\beta \rho_{\text{hel}}^\gamma \rho_{\text{air}}^\delta$ find value of β .

Space for Rough Work

2. An air filled parallel plate capacitor has capacitance $C = \frac{1}{2} \text{ F}$. Capacitor is connected through a resistor to a voltage source providing a constant potential difference $V = 2 \text{ volt}$.

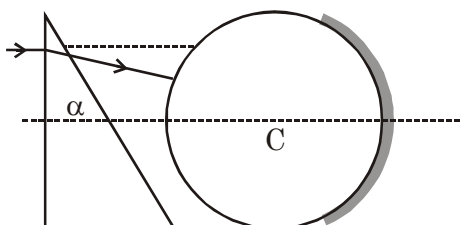


- A dielectric plate with dielectric constant $k = 2.5$ is inserted into the capacitor, filling it completely. After the equilibrium is established, the plate is quickly removed find total heat generated (in Joules), in resistor when the dielectric is removed.
3. A thin metal disc with mass $2.00 \times 10^{-3} \text{ kg}$ and radius 2 cm is attached at its center to a long fiber. The disk, when twisted and released, oscillate with a period of 1.00 s . The torsional constant of the fiber is $\delta \times 10^{-5} \text{ N.m/rad}$. Find value of δ (Take : $\pi^2 = 10$)

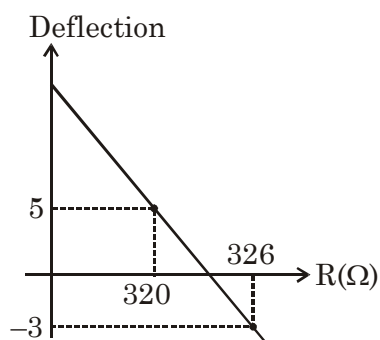


Space for Rough Work

4. A ray is incident normally on a right angle prism whose RI is $\sqrt{3}$ and prism angle $\alpha = 30^\circ$, after crossing prism ray passes through glass sphere. It strikes the glass sphere at $\frac{R}{\sqrt{3}}$ distance from principal axis, as shown in figure. Sphere is half polished. Find the angle of deviation of incident ray as multiplier of π .



5. In PO box, the graph of galvanometer deflection vs Resistance R (pulled out of resistance box) for the ratio 100 : 1 is given as shown (due to unsuitable values of R galvanometer shows deflection). The two consecutive values of R are shown in figure. The value of unknown resistance would be :



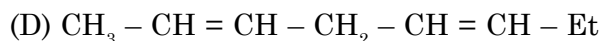
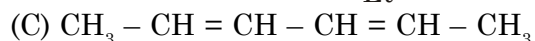
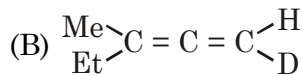
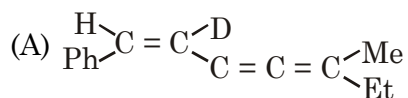
Space for Rough Work

PART-2 : CHEMISTRY

SECTION-I(i) : (Maximum Marks: 28)

- This section contains **SEVEN** questions.
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Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
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- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

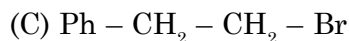
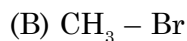
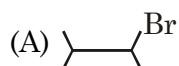
1. Which of the following compound(s) have four stereoisomers ?



2. $\text{Se}_3\text{O}_6\text{F}_3^{3-}$ is a symmetrical ion having cyclic structure with no peroxide linkage. Choose the correct option(s) regarding this species?
- (A) There are total five electron pairs around each Se atom.
 (B) This ion is aromatic
 (C) About each Se, shape is see-saw
 (D) The geometry about bridged oxygen is linear.

Space for Rough Work

3. Which of the following compound(s) can undergo both S_N2 & $E2$ reactions ?



4. Which of the following statement(s) is/are correct ?

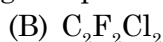
(A) Lyophilic sol get protected by lyophobic sol.

(B) Physisorption is irreversible process.

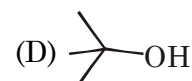
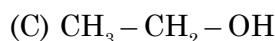
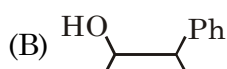
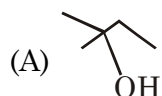
(C) When $FeCl_3$ solution is added to $NaOH$ solution, a negatively charged colloidal sol is obtained.

(D) In bredig's arc method, charge developed on colloidal particles is due to e^- capture.

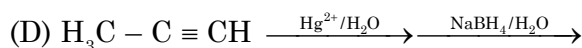
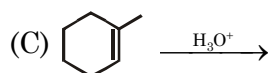
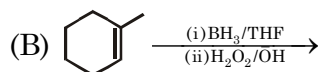
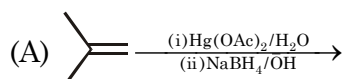
5. Which of the following compounds have both polar as well as non-polar structures?



6. Which of the following compound(s) can give positive dichromate test and iodoform test ?



7. Major alcohol product of which of the following reaction(s) give turbidity immediately in lucas test ?



Space for Rough Work

SECTION-I(ii) : (Maximum Marks : 18)

- This section contains **THREE** questions of matching type.
- This section contains **TWO** table (having 3 columns and 4 rows)
- Based on table, there are **THREE** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct
- For each question, darken the bubble corresponding to the correct option in the ORS.
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Full Marks : +3 If only the bubble corresponding to the correct option is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

Answer Q.8, Q.9 and Q.10 by appropriately matching the information given in the three columns of the following table.

Column-I	Column-II	Column-III
(I) $\Delta S_{\text{system}} = 0$	(i) $\Delta S_{\text{system}} = \Delta S_{\text{surrounding}}$	(P) $\Delta S_{\text{total}} = \Delta S_{\text{surrounding}}$
(II) $\Delta S_{\text{surrounding}} = 0$	(ii) $\Delta S_{\text{surrounding}} = n R \ln \left(\frac{V_2}{V_1} \right)$	(Q) $\Delta S_{\text{surrounding}} = +n R \ln \left(\frac{P_2}{P_1} \right)$
(III) $\Delta S_{\text{total}} = 0$	(iii) $\Delta S_{\text{system}} = n R \ln \left(\frac{V_2}{V_1} \right)$	(R) $\Delta S_{\text{total}} = n R \ln \left(\frac{P_1}{P_2} \right)$
(IV) $\Delta S_{\text{system}} = \Delta S_{\text{total}}$	(iv) $\Delta S_{\text{total}} = n R \ln \left(\frac{V_2}{V_1} \right)$	(S) $\Delta S_{\text{system}} = -n R \ln \left(\frac{V_1}{V_2} \right)$

[Given : V_1 = initial volume, V_2 = final volume, P_1 = initial pressure, P_2 = final pressure]

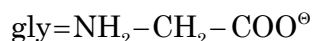
- Find correct option for reversible adiabatic expansion of fixed amount of an ideal gas?
 (A) (I)(iii)(R) (B) (II)(i)(P) (C) (II)(iii)(Q) (D) (III)(iii)(P)
- Find correct option for reversible isothermal expansion of fixed amount of an ideal gas?
 (A) (III)(iv)(R) (B) (II)(iv)(S) (C) (III)(iii)(Q) (D) (IV)(iii)(S)
- Which of the following option is correct for free expansion (Isothermal as well as adiabatic) of fixed amount of an ideal gas ?
 (A) (I)(iii)(P) (B) (IV)(iv)(R) (C) (III)(iii)(S) (D) (II)(iii)(Q)

Space for Rough Work

Answer Q.11, Q.12 and Q.13 by appropriately matching the information given in the three columns of the following table.

Columns-I, II and III contain complexes, isomerisms that can be exhibited, and number of stereoisomers of complexes (present in column - 1) respectively.

Column-I (Complexes)	Column-II Isomerism that can be shown	Column-III No. of stereoisomers
(I) $[\text{Co}(\text{NH}_3)_5(\text{NO}_2)]\text{Cl}_2$	(i) ionisation isomerism	(P) 0
(II) $[\text{Cr}(\text{NH}_3)_4\text{Cl}_2]\text{Br}$	(ii) Geometrical isomerism	(Q) 4
(III) $[\text{Cr}(\text{en})_2\text{Cl}_2]\text{Cl}$	(iii) Linkage isomerism	(R) 2
(IV) $[\text{Co}(\text{gly})_3]$	(iv) Optical isomerism	(S) 3



11. Choose the correct set of combinations:
 (A) (I), (iv), (P) (B) (II), (ii), (R) (C) (III), (iv), (Q) (D) (IV), (ii), (R)
12. Which of the following set is incorrect?
 (A) (I), (i), (P) (B) (II), (i), (R) (C) (III), (ii), (S) (D) (IV), (iv), (R)
13. Which complex can show all types of structural isomerism mentioned in column-2?
 (A) (I) (B) (II) (C) (III) (D) (IV)

Space for Rough Work

SECTION-II : (Maximum Marks : 15)

- This section contains **FIVE** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

+		-	
●	●	●	●
①	①	①	①
②	②	②	②
③	③	③	③
④	④	④	④
⑤	⑤	⑤	⑤
⑥	⑥	⑥	⑥
⑦	⑦	⑦	⑦
⑧	⑧	⑧	⑧
⑨	⑨	⑨	⑨

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

1. Molar conductivity at infinite dilution of NaCl, HCl and CH_3COONa are 130, 380 & $90 \Omega^{-1} \text{cm}^2 \text{mol}^{-1}$ respectively. Conductivity of $1.25 \times 10^{-3} \text{M}$ CH_3COOH is $3.4 \times 10^{-5} \Omega^{-1} \text{cm}^{-1}$. If $\alpha(\text{DOD})$ of CH_3COOH at 298 K is $y \times 10^{-2}$, then find y ?
2. The total number of angular nodes in all possible orbitals having $(n + l) \leq 6$ and $n > 2$, is

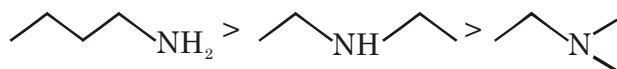
Space for Rough Work

3. 39.75 gram of Cu_2S is completely roasted with excess oxygen to produce CuO . The CuO thus obtained is now treated with excess of another sample of Cu_2S to produce a residue along with the evolution of SO_2 . The mass of elemental residue obtained gram is

(atomic weight of $\text{Cu} = 63.5$)

4. If x = number of correct statement(s) among the following

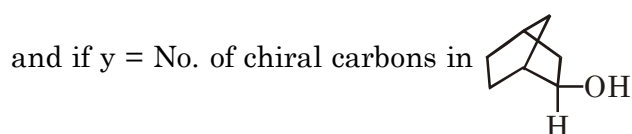
(i) Boiling point of



(ii) Nitrogen in $\text{NH}_2 - \text{NH}_2$ can be detected by Lassaigne's test.

(iii) A substance which decomposes below its boiling point can be best purified by distillation under reduced pressure.

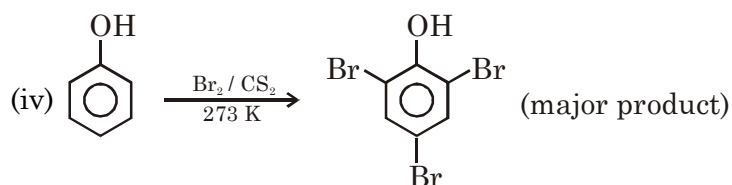
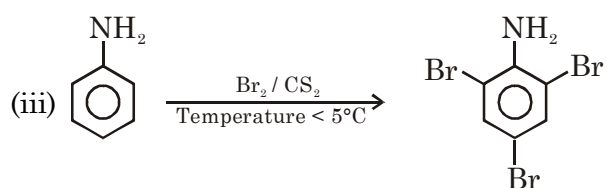
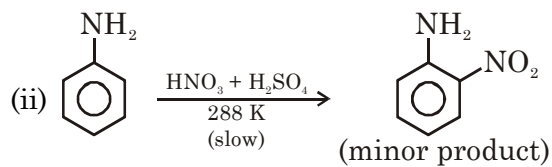
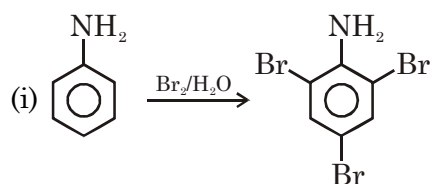
(iv) A mixture of camphor & benzoic acid can be separated by chemical methods



Then value of $(x^2 + y) = ?$

Space for Rough Work

5. If x = Number of correct reaction(s) among the following



The value of $(x^3 - 2)/2$ is

Space for Rough Work

PART-3 : MATHEMATICS**SECTION-I(i) : (Maximum Marks: 28)**

- This section contains **SEVEN** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -1 In all other cases.
- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

1. Let $y = y(x)$ be the solution of the differential equation

$$\frac{dy}{dx} + \left(\frac{2 \cos x - x \sin x}{x \cos x} \right) y = x^2 \sec x, y(1) = \sec 1, \text{ then}$$

(A) $\lim_{x \rightarrow \infty} (x^\alpha y \cos x) = \text{finite if } \alpha \in (-\infty, -3)$ (B) $\lim_{x \rightarrow \infty} (x^\alpha y \cos x) = \text{finite if } \alpha \in (-3, \infty)$

(C) $\left. \frac{dy}{dx} \right|_{x=1} = \frac{\tan 1 - 1}{\cos 1}$ (D) $\left. \frac{dy}{dx} \right|_{x=1} = \frac{\tan 1 + 1}{\cos 1}$

Space for Rough Work

2. Consider non zero vectors $\vec{a}, \vec{b}, \vec{c}$ and $\vec{p} = (\vec{a} \times \vec{b}) \times \vec{c}, \vec{q} = (\vec{b} \times \vec{c}) \times \vec{a}, \vec{r} = (\vec{c} \times \vec{a}) \times \vec{b}$ then

(A) $\vec{p} \cdot \vec{c} = 0$

(B) $\vec{p}, \vec{q}, \vec{r}$ can form a triangle

(C) $\vec{p}, \vec{q}, \vec{r}$ always form a Δ

(D) $\vec{p}, \vec{q}, \vec{r}$ are collinear

3. Consider the integral $I = \int_0^{\pi/6} \ln(\sin x) dx + \int_0^{\pi/6} \ln(4 \cos^2 x - 1) dx$, then

(A) $[I] = 0$

(B) $[I] = -1$

(C) $I < \int_0^{\pi/2} \ln(\sin x) dx$

(D) $I > \int_0^{\pi/2} \ln(\sin x) dx$

(where $[.]$ denotes greatest integer function)

4. Consider the function $f(x) = \int_0^{x^2} e^{t^2-t} dt$, then

(A) $f(x)$ has a local maxima at $x = 0$

(B) $f(x)$ has a local minima at $x = 0$

(C) $f(x) = 2$ has exactly two real solutions

(D) $f(x) = 2$ has exactly three real solutions

Space for Rough Work

5. If $f(x)$ is a thrice differentiable function, satisfying $f(x) = f'(x) \cdot f''(x)$ then (where ever exist)
- (A) $f'(x) \cdot f'''(x) = f'(x) - (f''(x))^2$ (B) $f(x) \cdot f'''(x) = f'(x) + (f''(x))^2$
- (C) $f(x) \cdot f'''(x) + (f''(x))^3 = f''(x)$ (D) $f(x) \cdot f'''(x) + (f''(x))^3 = f(x)$

6. If $\lim_{x \rightarrow 0} \frac{\sin\left(\frac{ax}{3}\right) + b - \tan\left(\frac{14x}{c}\right)}{x} = 3$, where $a, c \in \mathbb{N}$ and $b \in \mathbb{I}, c < 7$, then

- (A) Number of possible values of a is 4 (B) Number of possible values of a is 3
- (C) Number of possible values of b is 3 (D) Number of possible values of c is 4

7. Let $x = \sqrt{\frac{11 \dots 1}{2020 \text{ times } 1} - \frac{22 \dots 2}{1010 \text{ times } 2}}$. If the integer x can be expressed as $\frac{10^a - 1}{b}$, ($a, b \in \mathbb{N}$), then

- (A) $a + b$ is square of a natural number
- (B) $a + b$ is not the square of a natural number
- (C) Number of positive divisors of a is 8
- (D) Number of positive divisors of a is 6

Space for Rough Work

SECTION-I(ii) : (Maximum Marks : 18)

- This section contains **THREE** questions of matching type.
- This section contains **TWO** table (having 3 columns and 4 rows)
- Based on table, there are **THREE** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If only the bubble corresponding to the correct option is darkened.

Zero Marks : 0 If none of the bubbles is darkened.

Negative Marks : -1 In all other cases

Answer Q.8, Q.9 and Q.10 by appropriately matching the information given in the three columns of the following table.

If three normals at the points L, M, N on a parabola meet at (α, β) then

Column I	Column II	Column III
----------	-----------	------------

(P) For the parabola	(A) The centroid of the	(i) $\left(\frac{\alpha+4}{2}, \frac{\beta}{4}\right)$
----------------------	-------------------------	--

$y^2 = 4x$	ΔLMN is	
------------	-----------------	--

(Q) For the parabola	(B) The orthocenter of	(ii) $\left(\frac{2\alpha-4}{3}, 0\right)$
----------------------	------------------------	--

$y^2 = 8x$	ΔLMN is	
------------	-----------------	--

(R) For the parabola	(C) The circumcentre of	(iii) $\left(\frac{\alpha+8}{2}, -\frac{\beta}{2}\right)$
----------------------	-------------------------	---

$y^2 = 4x$	ΔLMN is	
------------	-----------------	--

(S) For the parabola	(D) The focus	(iv) $\left(\alpha-6, -\frac{\beta}{2}\right)$
----------------------	---------------	--

$y^2 = 8x$		
------------	--	--

8. Which of the following combination is correct?

(A) P, A, (ii)	(B) Q, A, (ii)	(C) R, C, (i)	(D) S, A, (ii)
----------------	----------------	---------------	----------------

9. Which of the following combination is correct?

(A) R, B, (i)	(B) R, B, (iv)	(C) S, B, (i)	(D) S, D, (iv)
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10. Which of the following combination is correct?

(A) P, A, (i)	(B) Q, B, (i)	(C) R, C, (i)	(D) S, C, (i)
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Space for Rough Work

Answer Q.11, Q.12 and Q.13 by appropriately matching the information given in the three columns of the following table.

Column I		Column II		Column III	
(P)	Number of 7 digit numbers	A	Which is/are of $a_1 a_2 a_3 a_4 a_5 a_6 a_7$ or $a_1 a_2 a_3 a_4 a_5$ type and $a_1 > a_2 > a_3 > a_4 > a_5 > a_6 > a_7$ or $a_1 > a_2 > a_3 > a_4 > a_5$ is	(i)	126
(Q)	Number of 5 digit numbers	B	Which doesn't change on reversing the order of digit is	(ii)	5/144
(R)	The probability that out of all 7 digit numbers if a numbers is randomly	C	Which has no two even or two odd digits consecutive & starts with an odd number	(iii)	36
(S)	The probability that out of all 5 digit numbers if a numbers is choosen	D	Which is/are of $a_1 a_2 a_3 a_4 a_5 a_6 a_7$ or $a_1 a_2 a_3 a_4 a_5$ type and $a_1 < a_2 < a_3 < a_4 < a_5 < a_6 < a_7$ or $a_1 < a_2 < a_3 < a_4 < a_5$	(iv)	5/576

On the basis of the above answer the following.

Note: Repetition of the digits is allowed for all the parts.

11. Which of the following combination is correct?
 (A) P, B, (i) (B) P, D, (iii) (C) Q, B, (i) (D) Q, D, (iii)
12. Which of the following combination is correct?
 (A) P, D, (i) (B) Q, D, (i) (C) R, D, (ii) (D) S, D, (ii)
13. Which of the following combination is correct?
 (A) P, C, (i) (B) R, C, (ii) (C) S, C, (iv) (D) S, C, (ii)

Space for Rough Work

SECTION-II : (Maximum Marks : 15)

- This section contains **FIVE** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darken the corresponding bubbles in the ORS.

For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

+									
●	●	0	0	●	0				
1	1	1	1	1	1				
2	2	2	2	2	●	2			
3	3	3	3	3	3				
4	4	4	4	4	4				
5	5	5	5	5	5	●			
6	6	6	6	6	6				
7	7	●	●	7	7				
8	8	8	8	8	8				
9	9	9	9	9	9				

-									
●	●	●	0	●	0				
1	1	1	1	1	1				
2	2	2	2	2	●	2			
3	3	3	3	3	3				
4	4	4	4	4	4				
5	5	5	●	5	5				
6	6	6	6	6	6				
7	7	7	7	7	7				
8	8	8	8	8	8				
9	9	9	9	9	9				

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

1. If $\int \left(1 + x \left(\cos x - \frac{\cos x}{\sin^2 x} \right) \right) e^{\sin x + \csc x} dx = f(x) + c$

(where $f\left(-\frac{\pi}{2}\right) = -\frac{\pi}{2}e^{-2}$, $f\left(\frac{\pi}{2}\right) = \frac{\pi}{a}e^b$ ($a, b \in \mathbb{N}$) then the value of $a + b$ is

2. Let $f(x) = e^{4x} - 8e^{2x} + 16x$ and $f^{-1}(x)$ be the inverse of function $f(x)$, which exists and it is differentiable. If $f'(x)$ denotes the derivative of $f(x)$, then the value of $(f^{-1})'(-7) + (f^{-1})(-7)$ is

Space for Rough Work

3. A bag contains two black and three white balls. Each time one ball is picked at random from the bag. If it is a white ball, a black ball is inserted in the bag, otherwise, a white ball is inserted in the bag. This process is continued until all balls in the bag are black. If the probability, it need not more than five picks to complete the process, is $\frac{a}{b}$ (a and b are co-prime), then $a + b$ is
4. Let X be a square matrix of order 2020, with real entries such that sum of elements of its each row is 1. If R_i denotes sum of elements of row number i of the matrix X^{2020} , then the value of $\frac{1}{5} \sum_{i=1}^{2020} R_i^2$ is
5. Number of possible integral values of 'a' for which the ellipse $x^2 + 4(y - a)^2 = 4$ intersects the parabola $x^2 = 2y$, is

Space for Rough Work

Space for Rough Work

QUESTION PAPER FORMAT AND MARKING SCHEME :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

17. Each part has two sections as detailed in the following table.

Section	Que. Type	No. of Que.	Category-wise Marks for Each Question				Maximum Marks of the section
			Full Marks	Partial Marks	Zero Marks	Negative Marks	
I(i)	One or more correct option(s)	7	+4 If only the bubble(s) corresponding to all the correct option(s) is(are) darkened	+1 For darkening a bubble corresponding to each correct option, provided NO incorrect option darkened	0 If none of the bubbles is darkened	-1 In all other cases	28
I(ii)	Matching Lists Type (Single correct option)	6	+3 If only the bubble corresponding to the correct option is darkened	—	0 If none of the bubbles is darkened	-1 In all other cases	18
II	Numerical Value Type (Up to second decimal place)	5	+3 If only the bubble corresponding to correct answer is darkened	—	0 In all other cases	—	15

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form number of the candidate, and that question paper and ORS codes are the same.

Signature of the Invigilator